MATH 241: Calculus III

Fall 2026

Lecture 20: Green's Theorem

Background & Motivation

Green's Theorem is the first of the three big theorems in vector calculus. It converts a line integral around a closed curve into a double integral over the enclosed region. This is powerful both computationally—sometimes one side is much easier than the other—and theoretically, as it connects circulation to curl.

1. Green's Theorem

– Converts a line integral around a closed curve into a double integral over the enclosed region (and vice versa)

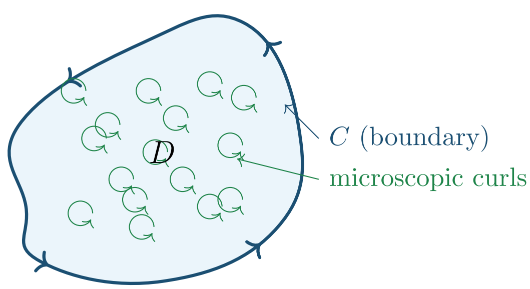
Theorem 1: Green's Theorem (Circulation Form)

Let C be a positively oriented (CCW), simple, closed curve bounding a region D. If P and Q have continuous partial derivatives on an open region containing D:

$$\boxed{\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA}$$

Requirements:

- C must be **simple, closed, positively oriented**
- P, Q must be **continuously differentiable on all of D**



Macroscopic circulation around C = sum of all microscopic curls $(Q_x - P_y)$ inside D.

Theorem 2: Green's Theorem (Vector Form)

Recall that $\oint_C P\,dx + Q\,dy$ is really a line integral of $\mathbf{F} = \langle P, Q\rangle$ along C. Writing Green's Theorem entirely in terms of $\mathbf{F}$:

$$\boxed{\oint_C \mathbf{F}\cdot d\mathbf{r} \;=\; \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt \;=\; \iint_D \left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dA}$$

The middle form is what we compute directly (parameterize C, dot, integrate). The right side is what Green's lets us swap to—a double integral over D.

Preview: The quantity $Q_x - P_y$ is what we will call the **curl** of $\mathbf{F}$ (next lecture). It measures local rotation. When we later learn **Stokes' Theorem**, we will see that Green's is just Stokes' with the surface flat in the xy-plane. Swap $Q_x - P_y$ for curl and you get Stokes'.

When to use Green's Theorem

- Hard line integral but easy double integral $\implies$ use Green's left-to-right
- Hard double integral but easy line integral $\implies$ use Green's right-to-left
- To compute areas via line integrals

Example 1: Verification on a Triangle (Both Sides)

Verify Green's Theorem for $\oint_C (x^2y\,dx + xy^2\,dy)$ around the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$.
Double integral side:
1. **Compute:** $P = x^2y$, $Q = xy^2$. $\quad Q_x - P_y = y^2 - x^2$
2. **Set up:** $\displaystyle\iint_D (y^2-x^2)\,dA = \int_0^1\int_0^{1-x}(y^2-x^2)\,dy\,dx$
3. **Inner integral:** $\displaystyle\left[\frac{y^3}{3}-x^2y\right]_0^{1-x} = \frac{(1-x)^3}{3} - x^2(1-x)$
4. **Evaluate:** $\displaystyle\int_0^1\left[\frac{(1-x)^3}{3}-x^2(1-x)\right]dx = \frac{1}{12}-\frac{1}{12} = 0$

Line integral side:
C_1: $(0,0)\to(1,0)$: $\mathbf{r} = \langle t, 0\rangle$, integral $= \int_0^1 0\,dt = 0$
C_2: $(1,0)\to(0,1)$: $\mathbf{r} = \langle 1-t, t\rangle$, $dx = -dt$, $dy = dt$
Integral $= \int_0^1[-(1-t)^2t + (1-t)t^2]\,dt = \int_0^1 t(1-t)(2t-1)\,dt$
$= \int_0^1(2t^3-3t^2+t)\,dt = \frac{1}{2} - 1 + \frac{1}{2} = 0$
C_3: $(0,1)\to(0,0)$: $\mathbf{r} = \langle 0, 1-t\rangle$, integral $= 0$
Total: $0+0+0 = 0$ ✓

Example 2: Using Green's to Simplify a Circle Integral

Evaluate $\displaystyle\oint_C (e^x+y^3)\,dx + (x^3-\sin y)\,dy$ where C is the unit circle (CCW).
1. **Identify:** $P = e^x + y^3$, $\quad Q = x^3 - \sin y$
2. **Compute:** $Q_x - P_y = 3x^2 - 3y^2$
3. **Convert to polar:** $\displaystyle\iint_D (3x^2-3y^2)\,dA = \int_0^{2\pi}\int_0^1 3r^2(\cos^2\theta - \sin^2\theta)\cdot r\,dr\,d\theta$
4. **Evaluate:** $\displaystyle= 3\int_0^1 r^3\,dr\cdot\int_0^{2\pi}\cos 2\theta\,d\theta = 3\cdot\frac{1}{4}\cdot 0 = \boxed{0}$

2. Area via Green's Theorem

Area Formulas via Green's Theorem

Setting $Q_x - P_y = 1$ in Green's Theorem gives area formulas. After parameterizing C with $\mathbf{r}(t) = \langle x(t), y(t)\rangle$:

$$\text{Area}(D) = \frac{1}{2}\int_a^b \left(x\,\frac{dy}{dt} - y\,\frac{dx}{dt}\right) dt$$

Also written in differential notation: $\frac{1}{2}\oint_C (x\,dy - y\,dx)$, or equivalently $\oint_C x\,dy$ or $-\oint_C y\,dx$.

Example 3: Area of an Ellipse

Find the area enclosed by $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ using Green's Theorem.

1. **Parameterize:** $x = a\cos t, \quad y = b\sin t, \quad t \in [0, 2\pi]$
2. **Compute differentials:** $dx = -a\sin t\,dt, \quad dy = b\cos t\,dt$
3. **Apply area formula:**

$$A = \frac{1}{2}\oint_C (x\,dy - y\,dx) = \frac{1}{2}\int_0^{2\pi} [a\cos t\cdot b\cos t - b\sin t\cdot(-a\sin t)]\,dt$$

4. **Simplify:** $= \dfrac{ab}{2}\displaystyle\int_0^{2\pi}(\cos^2 t + \sin^2 t)\,dt = \dfrac{ab}{2}\cdot 2\pi = \boxed{\pi ab}$

3. When Green's Theorem Fails

– Green's requires P, Q to have continuous partials on **all** of D

– If $\mathbf{F}$ is undefined inside D (e.g., at the origin), the theorem does **not** apply directly

Example 4: The $1/(x^2+y^2)$ Counterexample

Compute $\displaystyle\oint_C \frac{-y}{x^2+y^2}\,dx + \frac{x}{x^2+y^2}\,dy$ around the unit circle. Show Green's Theorem does not apply.

1. **Check partials:** $P = \dfrac{-y}{x^2+y^2}$, $Q = \dfrac{x}{x^2+y^2}$.

   $P_y = \dfrac{y^2-x^2}{(x^2+y^2)^2}, \quad Q_x = \dfrac{y^2-x^2}{(x^2+y^2)^2}. \quad$ So $Q_x - P_y = 0$!

   If Green's applied, the answer would be $\iint_D 0\,dA = 0$. But...
2. **Direct computation:** $\mathbf{r}(t) = \langle\cos t, \sin t\rangle$, $\mathbf{r}' = \langle -\sin t, \cos t\rangle$

   $\mathbf{F}\cdot\mathbf{r}' = (-\sin t)(-\sin t) + (\cos t)(\cos t) = 1$

   $\displaystyle\oint_C \mathbf{F}\cdot d\mathbf{r} = \int_0^{2\pi} 1\,dt = \boxed{2\pi} \neq 0$
3. **Why:** $\mathbf{F}$ is undefined at $(0,0)$, which lies inside C. The domain has a "hole," so Green's Theorem does not apply.

Practice Problems

Problem 1. Use Green's Theorem to evaluate $\oint_C (2y\,dx + 3x\,dy)$ around the unit square $[0,1]^2$ (CCW).

Answer: 1

Problem 2. Evaluate $\oint_C (y^2\,dx + x^2\,dy)$ where C is the boundary of the region between $y = x$ and $y = x^2$, $0 \le x \le 1$ (CCW).

Answer: $\frac{1}{30}$

Problem 3. Use Green's Theorem to find the area enclosed by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \le t \le 2\pi$.

Answer: $\frac{3\pi}{8}$

Problem 4. Evaluate $\oint_C (x^2 y\,dx - xy^2\,dy)$ where C is the circle $x^2 + y^2 = 4$ (CCW).

Answer: $Q_x - P_y = -y^2 - x^2 = -(x^2 + y^2)$; $\iint_D -(x^2 + y^2)\,dA = -\int_0^{2\pi}\int_0^2 r^3\,dr\,d\theta = -8\pi$

Problem 5. True or false (explain briefly):

(a) If $Q_x - P_y = 0$ everywhere in D, then $\oint_C P\,dx + Q\,dy = 0$.

(b) Green's Theorem can be applied to $\mathbf{F} = \langle x/r^2, y/r^2 \rangle$ (where $r = \sqrt{x^2 + y^2}$) on any region not containing the origin.

Answer: (a) True — by Green's Theorem, $\oint = \iint 0\,dA = 0$, provided P, Q are smooth on all of D.

(b) True — as long as D does not contain the origin, $\mathbf{F}$ is smooth on D and Green's applies.